

Steps to start **spring boot** project in java

Prerequisites

Before you begin, ensure you have the following installed:

- [Visual Studio Code](#)
- [Java Development Kit (JDK)](adoptium.net or other providers) version 17 or later
- [Maven](#) or Gradle build tool

Visual Studio Code +3

Step 1: Install Required Extensions

You need the correct extensions to enable Spring Boot development within VS Code. Visual Studio Code

1. Open the Extensions view (Ctrl+Shift+X or Cmd+Shift+X on macOS).
2. Install the [Spring Boot Extension Pack](#), which includes essential tools like the Spring Initializr, Spring Boot Tools, and the Spring Boot Dashboard.
3. Install the [Extension Pack for Java](#) to get basic Java support, debugging, and Maven integration. Visual Studio Code +2

Step 2: Create a New Project Using Spring Initializr

The Spring Initializr extension simplifies project creation.

Visual Studio Code

1. Open the **Command Palette** (Ctrl+Shift+P or Cmd+Shift+P on macOS).
2. Type Spring Initializr and select **Spring Initializr: Generate a Maven Project...** (or Gradle Project if you prefer Gradle).
3. Follow the prompts in the command palette:
 1. **Specify Spring Boot version:** Choose the latest stable version.
 2. **Specify project language:** Select Java.
 3. **Input Group Id:** Enter a group ID (e.g., com.example).
 4. **Input Artifact Id:** Enter an artifact ID for your project (e.g., demo).
 5. **Specify Packaging Type:** Choose Jar.
 6. **Specify Java version:** Select your installed JDK version (e.g., 17 or 21).
 7. **Search for dependencies:** Add necessary dependencies like Spring Web for web applications and Spring Boot DevTools for live reloading.
4. Select a folder on your computer where the project should be generated and click **Generate into this folder**.
5. VS Code will prompt you to open the newly created project. Click **Open**.

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Step 3: Run the Application

Once the project is loaded, VS Code will automatically detect and import the Maven or Gradle project.

1. Locate the main application file (e.g., DemoApplication.java) in the src/main/java folder.
2. Click the **Run** button that appears above the main method in the editor, or use the **Spring Boot Dashboard** in the sidebar to start the app.
3. The application will start, and the console output will show the embedded Tomcat server running, typically on <http://localhost:8080>.
4. Open a web browser and navigate to the address (e.g., <http://localhost:8080/hello> after adding a simple controller method) to see your running application.